TESLA COACHING CLASSES

Science and Technology – Part 2

Time: 2 Hours	Standard: Class 10	Total Marks: 40

Part Syllabus Test - 06 (PST-06)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 7	Introduction to Microbiology	86%
CHAPTER 1	Heredity and Evolution	3.5%
CHAPTER 2	Life Processes in Living Organisms Part – 1	3.5%
CHAPTER 3	Life Processes in Living Organisms Part – 2	3.5%
CHAPTER 6	Animal Classification	3.5%

Q.1(A) **Choose the correct alternative and rewrite the sentences:** **[5 Marks]**

i) _____ acid is used in the production of vitamins.

(a) Citric **(b)** Gluconic

(c) Lactic **(d)** Itaconic

ii) Yoghurt is a milk product produced with the help of _____.

(a) Lactobacilli **(b)** Azotobacter

(c) Corynebacterium **(d)** Streptococcus

iii) _____ is a powerful antibiotic against treatment of tuberculosis.

(a) Penicillin **(b)** Rifamycin

(c) Streptomycin **(d)** Bacitracin

iv) _____ is used in the commercial bakery industry.

(a) Compressed yeast **(b)** Algae

(c) Bacteria **(d)** Microbes

v) _____ is a substance obtained by microbial processing that functions as an artificial sweetener.

(a) Nycin **(b)** Lysine

(c) Xanthenes **(d)** Xylitol

Q.1(B) **Answer the following sub-questions:** **[5 Marks]**

i) Find the odd one out:

(a) Penicillin	**(b)** Neomycin
(c) Isomerase	**(d)** Rifamycin

ii) Identify the correlation and rewrite:
Theory of evolution : Darwin : : _____ : Lamarck

iii) Name the following:
An enzyme obtained from fungi to produce vegetarian cheese.

iv) State whether True or False:
Methane gas is produced by microbial oxidation of industrial waste.

v) Answer in one sentence:
Write any one cold-blooded animal you know.

Q.2(A) **Give scientific reasons (Any TWO):** **[4 Marks]**

i) Bread and other products made using commercial yeast are nutritious.

ii) Microorganisms are used to control the effects of sea oil spills.

iii) The landfill pit is lined with plastic sheets.

Q.2(B) **Answer the following questions (Any THREE):** **[6 Marks]**

i) Write a short note: Bioinsecticides

ii) How is soil pollution controlled by microbes?

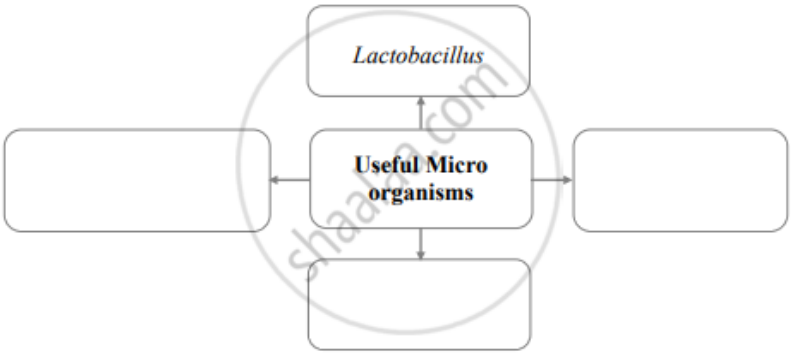
iii) Why is water an essential nutrient?

iv) Which fuels do we get from microbes?

v) What is the use of antibiotics to human and other animals?

Q.3 **Answer the following questions (Any FIVE):** **[15 Marks]**

i) Complete the concept map.



ii) Complete the paragraph by selecting the appropriate option in parentheses.

(energy, microbial, erosion-proof, reused, purification, environmentally friendly)

Nowadays, instead of chemical catalysts, _____ enzymes are used in the chemical industry. These enzymes are active at low temperature, pH and pressure; due to which _____ is saved and _____ instruments are also not necessary. Enzymes carry out specific processes; hence unnecessary by-products are not formed due to which expenses on _____ are minimised. In case of microbial enzymatic reactions elimination and decomposition of waste material is avoided and enzymes can be _____ Hence, such enzymes are _____.

iii) What products do you get from microorganisms through the fermentation process? Explain with an example.

iv) Write the answers to the questions by observing the figure.

a. What type of fuel production process is shown in the figure?

b. Write two examples of each of the solid, liquid and gaseous fuels produced in this fuel production process?

c. How do microorganisms play their role in this process?



v) Explain – Importance of microbes in sewage (water) treatment in cities.

vi) How are microbes useful in land filling for solid waste management?

vii) What are the benefits of microbial inoculation in plants?

viii) Explain two main process in sexual reproduction.

Q.4	Answer the following question (Any ONE):	[5 Marks]

i) Read the following paragraph and write the answers to the questions asked.

Sulphuric acid is present in the acid rain and materials coming out of mines. You know that erosion of metals present in statues, bridges and buildings occurs due to it. Sulphuric acid is a source of energy for some species of bacteria like Acidiphilium spp. and Acidithiobacillus ferrooxidans. Hence, these bacteria can control the soil pollution occurring due to acid rain. Water soluble salts of uranium are present in the wastes produced during electroplating and in effluent released in the environment from the atomic energy plant. Geobacter convert these salts of uranium into insoluble salts and thereby prevent those salts from mixing with groundwater sources.

a. What causes metal erosion in statues, bridges and buildings?

b. Sulphuric acid is the source of energy for which bacteria?

c. What kind of pollution do these bacteria control?

d. What are the water soluble salts in atomic energy plants and in the process of electroplating?

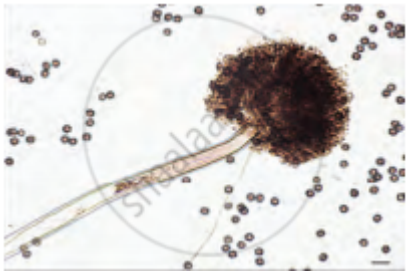
e. Which bacteria prevent these salts from mixing with ground water by converting them Into insoluble salts?

ii) Observe the figure and write the answers to the questions asked.

a. Write the name of the fungus in the figure above.

b. What is the source of these fungi?

c. Which organic acid obtained from this organism is used in commercial production?



ALL THE BEST

DESIGNED BY YOUR MENTOR – LUCKY BHAIYA